these M.2 SATA SSDs are limited to 600 MBps and current gen M.2 SATA SSDs barely heat up. However, the newer NVMe protocol allows for much higher transfer speeds. An NVMe M.2 SSD running at 3,600 MBps is a common sight these days. Incidentally, the higher speeds also mean that these newer NVMe SSDs heat up quite a bit under constant load.


Intel 750 Series water block


Passive heatsink


HDD water cooling block

The key word here is "constant". With storage media, constant load is achieved if you're transferring files to and from the SSD to the extent that it is running at max bandwidth all the time. You would have guessed that this is a rarity. In fact, a high-end NVMe SSD will run at 45 degree celsius or so under normal use and with constant load, that can rise up to 55-60 degrees celsius. This is well within the normal functioning temperatures for these NVMe SSDs. They can get hotter if the nearby components are radiating too much heat or if the air-flow within the case is insufficient. Most motherboards ship with multiple NVMe slots with the primary slot being very close to the CPU. In such cases, you might experience a bit of extra heat.

Then there is SSD overclocking so you know where this is going. If normal temperatures are in the 45-60 degree range then overclocking your SSD will push it way outside the comfort zone. For that you have normal SSD heatsinks as well as water blocks.

Hard drives

Hard drives tend to get hot on their own but they aren't as susceptible to heat damage because a huge chunk of your average hard drive is just solid